

Booklet No.

Civil Engineering Paper-I

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional

QUESTION PAPER SPECIFIC INSTRUCTIONS

Please read each of the following instructions carefully before attempting questions :

1. There are **EIGHT** questions divided in two Sections, out of which **FIVE** are to be attempted.
2. Question no. 1 and 5 are compulsory. Out of the remaining questions, **THREE** are to be attempted choosing at least **ONE** question from each Sections.
3. The number of marks carried by a question/sub question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note – Candidates will be allowed to use Scientific (Non-programmable type) calculators.

SECTION - A

1. Answer **any five** of the following :

(8x5=40)

- (a) A beam ABCD, 5 m long hinged at A and roller supported at D, is subjected to a point load of 10 kN at B and a clockwise moment at C as shown in Fig. 1.1. If $EI = 1.2 \times 10^3 \text{ kNm}^2$, determine deflection at point B.

(8)

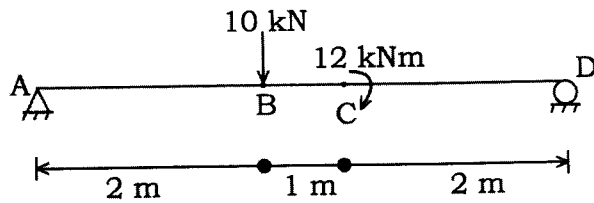
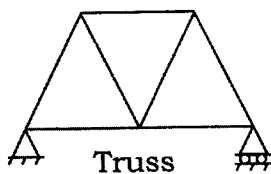


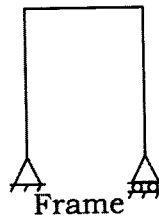
Fig. 1.1

- (b) Find degree of static indeterminacy (DOSI) for following structures.

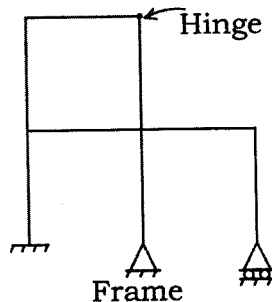
(8)



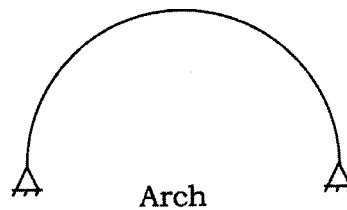
Truss



Frame



Frame



Arch

- (c) Find value of 'x' for given equation subjected to tolerance of error = ± 0.5 by bisection method.

(8)

$$x^3 - 4x - 7 = 0$$

- (d) Derive stiffness matrix of a two noded beam member having rotation and vertical displacement as degree of freedom at each end. Draw necessary sketches also for derivation.

(8)

- (e) Derive total collapse load for following cases. Also state difference between total collapse loads of both cases.

(8)

- Simply supported beam carrying UDL over entire length.
- Both end fixed beam carrying UDL over entire length.

- (f) Write causes of accidents on construction site. Also write role and responsibility of following : (8)
- employee
 - site engineer
 - safety officer/manager
- (g) A tension member is formed by gluing two wooden scantlings each $100 \text{ mm} \times 100 \text{ mm}$ at their ends which are cut at 60° as shown in Fig. 1.2. Calculate the safe value of pull P if the permissible normal and shear stress in the glue are 2 N/mm^2 and 1.5 N/mm^2 . (8)

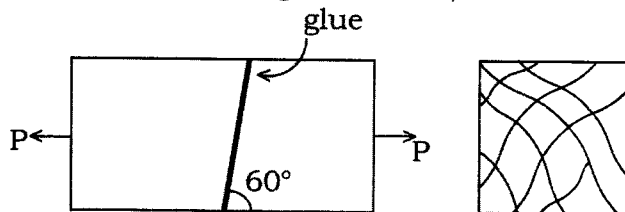


Fig. 1.2

2. (a) A beam ABCDEFG, 16 m long, has hinged support at G and roller supports at A & D. The beam has internal hinge at point C. The beam carries three vertical loads as shown in Fig. 2.1. Calculate (i) support reactions at A, D and F (ii) Draw SFD and BMD (iii) locate point of contraflexure. (15)

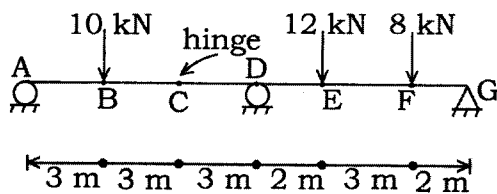


Fig. 2.1

- (b) A beam ABC has $AB = 5 \text{ m}$, $BC = 3 \text{ m}$. Ends A and C are fixed and intermediate support B is hinged support. The loading on beam is as shown in Fig. 2.2. Draw SFD and BMD of the beam using slope deflection method. (15)

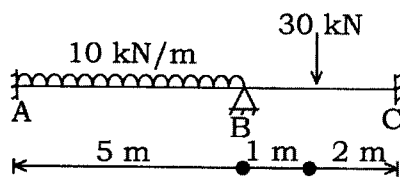


Fig. 2.2

- (c) Solve these simultaneous equations using Gauss elimination method. Also comment on difference between Gauss elimination and Gauss Jordan method. (10)

$$8x + 5y + 3z = 37$$

$$4x + 10y + 3z = 35$$

$$2x + 4y + 5z = 19$$

3. (a) A simply supported beam of 16 m span is subjected to a uniform dead load of 5000 N/m and a uniformly live load (longer than span) of 8000 N/m. Determine (i) maximum and minimum shear force at left hand quarter point (at point 4 m from left support) (ii) the maximum BM at the same point.

(15)

(iii) the range over which shear force may have +Ve or -Ve values.

- (b) A bracket is connected with ISMB650 as shown in Fig. 3.1. Enlist possible types of failures in this connection. Explain each failure with illustrative figures.

(15)

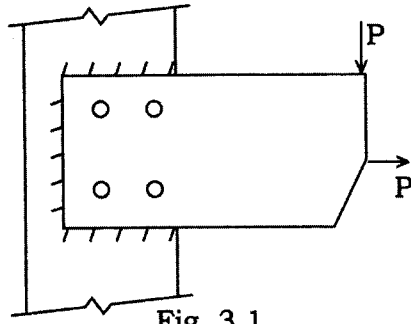


Fig. 3.1

- (c) Enlist ten different advanced construction machineries and explain their component parts, working and uses.

(10)

4. (a) A small project is composed of seven activities as given in Table 4.1 (a) (i) draw project network (ii) Find expected duration and variance of all activities (iii) What is expected project length ? (iv) Calculate the variance of project duration (v) What is probability that project will be completed at least 3 weeks earlier than expected ?

(15)

Table 4.1 (a)

Activity		Estimated Duration (Week)		
i	j	Optimistic	Most likely	Pessimistic
1	2	1	1	7
1	3	1	4	7
1	4	2	2	8
2	5	1	1	1
3	5	2	5	14
4	6	2	5	8
5	6	3	6	15

- (b) A continuous beam ABCDE has lengths $AB=3$ m, $BC=6$ m, $CD=3$ m and $DE=3$ m. The beam carries three downward point loads and one UDL as shown in Fig. 4.1 (b). The portion AB is overhanged. Supports at B, C, D are roller supports and support of E is hinged. For entire beam EI is constant. Determine support moments for this continuous beam by moment distribution method. (15)

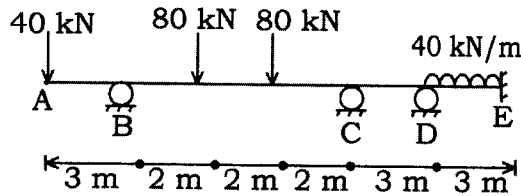


Fig. 4.1 (b)

- (c) A fixed beam ABC is clamped at A & C. Flexure rigidity of AB and BC parts are EI and 2EI respectively. The length of parts $AB=3$ m and $BC=3$ m. The entire beam is loaded with uniformly distributed load of 40 kN/m as shown in Fig. 4.1 (c). (10)

Draw BMD of the fixed beam.

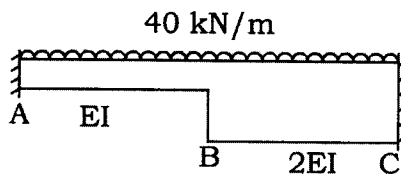


Fig. 4.1 (c)

SECTION - B

5. Answer **any five** of the following : (8x5=40)
- (a) Derive the governing equation of doubly reinforced concrete beam. (8)
 - (b) (i) State various types of structural systems with reference to multistoreyed building and illustrate them by sketches. (4)
 - (ii) Explain load carrying mechanism and structural action of any two of the systems. (4)
 - (c) (i) State and explain the factors influencing the selection of type of bridge. (4)
 - (ii) Define impact factor with reference to design of bridges. Explain its importance in the design of bridges. (4)

- (d) Describe in detail the test on fresh concrete accordance with relevant IS code. State its relevance in practice and give explanation for the same. (8)
- (e) Give classification of prestressing systems in detail. Explain, in detail, any one of the posttensioning systems commonly adopted in the field. (8)
- (f) A circular area 8 m diameter carries a UDL of 100 kN/m^2 . Determine the vertical stress at a depth of 2 m, 4 m and 8 m. Plot the variation of vertical stress with depth. (8)
- (g) (i) State the requirements for good formwork for construction purposes. Also, describe about the material used and criteria stipulated for various types of formwork. (4)
- (ii) Describe in detail about slipform formwork system used in construction activity along with its applications. (4)
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6. (a) Design a strap footing combined foundation for two columns 'A' and 'B' spaced 6 m apart centre to centre. Column 'A' is 400 mm square and carries a service load of 500 kN. Column 'B' is 500 mm square and supports a service load of 1000 kN. The safe bearing capacity of the soil at the site of footing is 200 kN/m^2 . Adopt M20 grade concrete and Fe415 grade steel (HYSD) sketch the reinforcement details. (15)
- (b) A cantilever type retaining wall is to be built to retain earth level with the top of the wall at a height of 5.5 m above ground level. The density of the soil is 18 kN/m^3 with a safe bearing capacity of 130 kN/m^2 . Design the retaining wall assuming the shearing resistance of the soil as 35° . Take coefficient of friction between soil and concrete as 0.56. Adopt M20 grade concrete with Fe 415 HYSD bars as reinforcement. (15)
- (c) A T-beam floor span of reinforced concrete has a slab of 160 mm thick spanning between T-beams which are placed 3.3 m apart. The T-beams have a clear span of 8.0 m and the end bearing are 400 mm thick walls. The live load on the floor is 4 kN/m^2 . Use M20 grade concrete and Fe415 HYSD bars to design the intermediate T-beam. Sketch the cross-section and longitudinal section of the beam showing reinforcement details. Use IS456-2000 Codal provisions for design. (10)
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7. (a) (i) What are the different components of a bridge ? Explain it with neat sketches. (5)
- (ii) What is meant by economical span of the bridge ? Derive the equation for economical span. (2+4)
- (iii) State different type of piers usually used in the construction of Highway bridges. (4)
- (b) (i) What are different stages of manufacturing concrete ? Describe in detail mixing, placing and transportation of concrete. (9)
- (ii) State and explain principles and objectives of concrete mix design. (6)
- (c) A stratum of clay with an average liquid limit of 45% is 6 m thick. Its surface is located at a depth of 8 m below the ground surface. The natural water content of the clay is 40% and specific gravity is 2.7. Between ground surface and clay, the subsoil consist of fine sand. The water table is located at a depth of 4 m below the ground surface. The average submerged unit weight of sand is 10.5 kN/m^3 and the unit weight of sand above the water table is 17 kN/m^3 . The weight of building that will be constructed on the sand above clay increases the overburden pressure on the clay by 40 kN/m^2 . Estimate the settlement of building. (10)
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8. (a) A PSC beam supports a liveload of 5 kN/m over a simply supported span of 8.0 m. The beam is to be prestressed by an effective prestressing force of 250 kN at a suitable eccentricity such that the resultant stress at the soffit of the beam at midspan section is zero. The cross-section of the beam is symmetrical I-section having overall depth of 420 mm with thickness of the flange and web being 60 mm and 80 mm respectively. Width of the flange is 200 mm. (15)
- (i) Find the eccentricity of the PSF.

- (ii) If the tendon is concentric, what should be magnitude of PSF for the resultant stress to be zero at the bottom fibre at the midspan section.
- (b) In a direct shear test conducted on a dense sand, the sample fails at a shear stress of 75 kN/m^2 , when the normal stress was held constant at 100 kN/m^2 . Draw the Mohr's circle for the failure condition and determine, (i) angle of shearing resistance, (ii) the orientation of major and minor principal plane and the stresses acting on them, and (iii) the orientation of the plane of maximum shear stress. If a specimen of this soil were to be tested in a triaxial shear test under CD condition at a cell pressure of 125 kN/m^2 , determine the axial stress at which sample would fail. (15)
- (c) (i) Why is high tensile steel needed for prestressed concrete construction. (5)
- (ii) What is anchorage zone ? Explain about the anchorage zone stresses with the help of neat sketches. (5)

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